CV802 Sub-project 2 — Surface Extraction (one page)

Goal. Turn the coloured 3D points of Sub-project 1 into a textured triangle mesh (`.obj`), plus a mesh viewer. **Inputs.** Light-shirt subject: (a) VGGSfM cloud after person-mask cleanup, 138 k points; (b) COLMAP MVS cloud, 487 k points with normals. **Pipeline.** Method → `geometry.ply` → shared finalizer: UV atlas, texture baked from the input colours (or the method's own colour), textured OBJ, metrics vs the input (Chamfer as % of the bbox diagonal, F-score at 1 % / 0.5 %, normal consistency, colour PSNR), renders. Everything ran on one RTX 5000 Ada through Slurm; a browser UI runs methods, shows progress/GPU, and views/compares meshes (three.js).

Methods. (1) **Screened Poisson, implemented from scratch** (no API): PCA+MST normals, density-weighted normal splatting, screened Poisson system solved by PCG on a GPU grid (coarse-to-fine), own marching tetrahedra, density trimming. (2) **IER meshing** (ECCV'20, pretrained): candidate triangles scored by intrinsic-extrinsic ratio. (3) **Point2Mesh** (SIGGRAPH'20): self-prior MeshCNN shrink-wraps a convex hull. (4) **MeshAnything V2** (2024, pretrained): autoregressive artist mesh, fed a dense mesh via the authors' `--mc` path. (5) **Deep Hybrid Self-Prior** (ICCV'21): 3D + 2D self-priors for shape and colour. (6) Extra: **NKSR** (CVPR'23, pretrained neural kernel field).

Best run	VGGSfM Chamfer % / F@1%	MVS Chamfer % / F@1%	Faces	Watertight
Poisson (ours)	**0.149 / 0.999**	0.100 / 0.999	0.3–1 M	optional (untrimmed)
NKSR (extra)	0.156 / 0.997	**0.080 / 1.000**	2–3 M	no
IER	0.187 / 0.998	0.120 / 1.000	17 k	no
DHSP 3D prior	0.162 / 0.999 (crumpled)	0.152 / 0.980	16 k	yes
MeshAnything V2	0.295 / 0.987	0.518 / 0.873	1–1.4 k	no
Point2Mesh	0.553 / 0.844	3.98 / 0.133	10 k	yes

Observations.

- A carefully implemented classical method (screened Poisson) matches pretrained networks on dense, clean clouds; NKSR is best on the denser MVS cloud and reconstructs in seconds.
- Input domain matters: MeshAnything went from F@1% 0.51 (raw points) to 0.987 with the authors' dense-mesh preprocessing.
- Self-prior methods are fragile on real scans: Point2Mesh depends on normal orientation (`--unoriented`: 0.65 → 0.84) and breaks on outliers; DHSP's 3D prior (Chamfer + 0.2·edge, as in the paper) is crumpled, and its 2D refinement needs a continuous UV atlas that OptCuts did not deliver in practical time.
- Point-to-surface metrics hide surface quality — normal consistency and shaded renders exposed DHSP's crumpling.



Poisson (ours) | NKSR | IER | DHSP | MeshAnything V2 | Point2Mesh

Front renders of the best run per method on the MVS input (texture from the finalizer or the method).

Deliverables. Textured OBJ for every method and input; mesh viewer UI with side-by-side comparison; 3D-print STLs (180 mm, watertight checks). Recommended for printing: MVS screened-Poisson watertight mesh.